

WEB DEVELOPMENT

Q.NO = 1

```
#include <iostream>
#include <vector>
using namespace std;

class Solution {
public:
    vector<vector<string>> ans;

    bool isSafe(int row, int col, vector<string> &board, int n) {

        for(int i = row - 1; i >= 0; i--) {
            if(board[i][col] == 'Q')
                return false;
        }

        for(int i = row - 1, j = col - 1; i >= 0 && j >= 0; i--, j--) {
            if(board[i][j] == 'Q')
                return false;
        }

        for(int i = row - 1, j = col + 1; i >= 0 && j < n; i--, j++) {
            if(board[i][j] == 'Q')
                return false;
        }

        return true;
    }

    void solve(int row, vector<string> &board, int n) {

        if(row == n) {
            ans.push_back(board);
            return;
        }

        for(int col = 0; col < n; col++) {

            if(isSafe(row, col, board, n)) {

                board[row][col] = 'Q';
```

```

        solve(row + 1, board, n);
    }
    board[row][col] = '.';
}
}

vector<vector<string>> solveNQueens(int n) {
    vector<string> board(n, string(n, '.'));
    solve(0, board, n);
    return ans;
}
};

int main() {
    int n;
    cin >> n;

    Solution obj;

    vector<vector<string>> result = obj.solveNQueens(n);
    cout << "Total Solutions = " << result.size() << endl;

    for(int i = 0; i < result.size(); i++) {
        cout << "\nSolution " << i + 1 << ":\n";
        for(string row : result[i]) {
            cout << row << endl;
        }
    }

    return 0;
}

```

OUTPUT:-

```
4
Total Solutions = 2

Solution 1:
.Q..
...Q
Q...
..Q.

Solution 2:
..Q.
Q...
...Q
.Q..

...Program finished with exit code 0
Press ENTER to exit console.□
```

Q.NO = 02

```

#include <stdio.h>
#include <string.h>

int match(char str[], char pattern[]) {
    int i = 0, j = 0;

    while(str[i] != '\0' && pattern[j] != '\0') {

        // normal character match
        if(pattern[j + 1] == '+') {

            if(str[i] != pattern[j])
                return 0;

            // one or more same characters
            while(str[i] == pattern[j]) {
                i++;
            }

            j = j + 2;
        }
        else {

            if(str[i] != pattern[j])
                return 0;

            i++;
            j++;
        }
    }

    // both should end together
    if(str[i] == '\0' && pattern[j] == '\0')
        return 1;

    return 0;
}

int validPattern(char pattern[]) {
    int n = strlen(pattern);

```

```

    // '+' cannot come first
    if(pattern[0] == '+')
        return 0;

    // two '+' together not allowed
    for(int i = 0; i < n - 1; i++) {

        if(pattern[i] == '+' && pattern[i + 1] == '+')
            return 0;
    }

    return 1;
}

int main() {

    char P[100], Q[100];

    printf("Enter String P : ");
    scanf("%s", P);

    printf("Enter Pattern Q : ");
    scanf("%s", Q);

    if(validPattern(Q) == 0) {

        printf("Invalid Pattern");
        return 0;
    }

    if(match(P, Q))
        printf("True");
    else
        printf("False");

    return 0;
}

```

OUTPUTS :-

```
Enter String P : AAAB  
Enter Pattern Q : A+B  
True
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```

```
Enter String P : AABA  
Enter Pattern Q : BA  
False
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```